GNANA MATH • NUMERICAL METHODS OF INTEGRATION

Page 1 — Formula Reference | Definitions • Formulae • Properties • Conditions • Quick Examples

■ 1. Fundamental Definitions

Definite Integral (to be approximated): $I = \int_a^b f(x)\,dx$

When an antiderivative F(x) is unavailable or difficult, we replace the integral by a weighted sum of function values.

Partition: Divide [a, b] into n subintervals of width $h = (b - a)/n$. Nodes: $x_i = a + i\,h$, $i = 0,1,\ldots,n$.

Error: $E = I - I_{approx}$. Order of a method is the highest power of h that appears in the leading error term.

■ 2. Important Formulae (Newton–Cotes Family)

Trapezoidal Rule (Composite)

$$I \approx (h/2)\,[y_0 + 2(y_1+\ldots+y_{n-1}) + y_n]$$

Local error (one panel): $-(b-a)^3 f''(\xi)/12$ | Global error: $-(b-a)h^2 f''(\xi)/12 \rightarrow O(h^2)$

Simpson's 1/3 Rule (Composite) — n must be even

$$I \approx (h/3)\,[y_0 + 4(y_1+y_3+\ldots) + 2(y_2+y_4+\ldots) + y_n]$$

Global error: $-(b-a)h^4 f^{(4)}(\xi)/180 \rightarrow O(h^4)$

Simpson's 3/8 Rule (Composite) — n multiple of 3

$$I \approx (3h/8)\,[y_0 + 3(y_1+y_2+y_4+y_5+\ldots) + 2(y_3+y_6+\ldots) + y_n]$$

Global error: $-(b-a)h^4 f^{(4)}(\xi)/80 \rightarrow O(h^4)$

Midpoint (Rectangle) Rule (Composite)

$$I \approx h\,[f(x_0+h/2) + f(x_1+h/2) + \ldots + f(x_{n-1}+h/2)]$$

Global error: $(b-a)h^2 f''(\xi)/24 \rightarrow O(h^2)$ (same order as Trapezoidal, often smaller constant)

■ 3. Properties and Identities

- **Linearity**: Numerical integration operators are linear → can integrate term-by-term.
- **Degree of precision**: Trapezoidal & Midpoint exact for polynomials ≤ degree 1; Simpson exact ≤ degree 3.
- **Romberg Integration**: Richardson extrapolation applied to Trapezoidal rule → successively higher-order methods.
- **Gaussian Quadrature**: Optimal nodes & weights → n-point rule exact for degree ≤ 2n–1 (e.g., Gauss–Legendre).
- **Composite vs Single**: Composite formulas reduce global error by applying a basic rule on many small panels.
- **Error ratio**: For Trapezoidal, halving h reduces error by ≈ 4; for Simpson by ≈ 16 (when asymptotic regime is reached).

■ 4. Conditions of Applicability

- f must be continuous on [a, b] (or integrable singularities treated carefully).
- For error formulas involving f″ or f■■■ the corresponding derivative must exist and be continuous.
- Simpson 1/3 requires **even** number of subintervals (n even).
- Simpson 3/8 requires n multiple of 3. Mixed 1/3 + 3/8 can handle any n ≥ 1.
- Open formulas (Midpoint, Gauss) do not evaluate endpoints → useful when f is singular at a or b.
- Adaptive methods (not detailed here) should be preferred when f varies sharply or has singularities.

■ 5. Quick Reference Examples

Example A — Trapezoidal, n=4, $\int_{■}^{1} e^x\,dx$ (exact = e–1 ≈ 1.71828)

h=0.25; y=[1, 1.2840, 1.6487, 2.1170, 2.7183] → I ≈ (0.25/2)[1 + 2(1.2840+1.6487+2.1170) + 2.7183] ≈ 1.7272 (err ≈ 0.0089)

Example B — Simpson 1/3, same data, n=4 (even)

I ≈ (0.25/3)[1 + 4(1.2840+2.1170) + 2(1.6487) + 2.7183] ≈ 1.71832 (err ≈ 4×10^{-5})

Rule of thumb: Prefer Simpson when n even and f is smooth; use Trapezoidal for quick estimates or when only endpoint data are reliable.

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Page 2 — Methods & Solution Procedures | Selection Guide • Algorithms • Worked Examples • Mistakes • Verification

■ 1. Method Selection Guide

- **Quick estimate / low accuracy needed** → Trapezoidal or Midpoint (simple, robust).
- **Smooth f, moderate–high accuracy, even n** → Composite Simpson 1/3 (best accuracy per evaluation among basic rules).
- **n multiple of 3** → Simpson 3/8 (or combine 1/3 + 3/8 for arbitrary n).
- **f singular or unknown at endpoints** → Midpoint or open Newton–Cotes / Gauss quadrature.
- **Very high accuracy, smooth f** → Romberg integration or adaptive Gaussian quadrature.
- **Only a few function evaluations allowed** → Gaussian quadrature (highest precision for given n).

■ 2. Step-by-Step Solution Algorithms

Algorithm A — Composite Trapezoidal / Simpson

1. Choose n (even for Simpson 1/3). Compute h = (b–a)/n.
2. Generate nodes $x_i = a + i\,h$ and evaluate $y_i = f(x_i)$ for i = 0…n.
3. Apply the appropriate weighted sum (see Page 1 formulae).
4. (Optional) Estimate error using the analytic error formula if a bound on the relevant derivative is known, or by repeating with 2n and comparing.

Algorithm B — Romberg Integration (outline)

1. Compute Trapezoidal approximations T_k for successive halvings of h (k = 0,1,2,…).
2. Build the Romberg table by Richardson extrapolation:

$$R_{k,j} = (4^j R_{k,j-1} - R_{k-1,j-1}) / (4^j - 1)$$

3. Stop when consecutive diagonal entries agree to the required tolerance.

■ 3. Worked Examples

Worked Example 1 — Simpson 1/3 on $\int_0^{\pi/2} \sin x\,dx$ (exact = 1)

n = 4 (even), h = π/8 ≈ 0.3927

x = [0, π/8, π/4, 3π/8, π/2]

y ≈ [0, 0.3827, 0.7071, 0.9239, 1.0000]

I ≈ (h/3)[0 + 4(0.3827+0.9239) + 2(0.7071) + 1] ≈ 1.00013 (excellent agreement)

Worked Example 2 — Trapezoidal vs Simpson on $\int_1^3 (1/x)\,dx$ (exact = ln 3 ≈ 1.0986)

n = 4, h = 0.5; y = [1, 0.6667, 0.5, 0.4, 0.3333]

Trapezoidal: (0.5/2)[1 + 2(0.6667+0.5+0.4) + 0.3333] ≈ 1.1167 (err ≈ 0.018)

Simpson 1/3: (0.5/3)[1 + 4(0.6667+0.4) + 2(0.5) + 0.3333] ≈ 1.1000 (err ≈ 0.0014)

→ Simpson is an order of magnitude more accurate for the same number of evaluations.

■ 4. Common Mistakes

- Using Simpson 1/3 with an **odd** number of intervals → formula is invalid; switch to 3/8 or mix rules.
- Forgetting the factor of 2 (or 4) on interior points → systematically under- or over-weights the sum.
- Applying the single-panel error formula to a composite rule without multiplying by the number of panels.
- Assuming higher-order methods always win: if f has large high derivatives (or is only C^1), Trapezoidal may be safer.
- Ignoring units / scaling of h when comparing errors across different intervals.
- Evaluating f at a singularity (e.g., 1/√x at 0) with a closed rule → use an open rule or a substitution first.

■ 5. Verification Techniques

- **Known exact value**: Compare against an analytic integral when available (as in the examples).
- **Doubling n**: Compute I(h) and I(h/2). For Trapezoidal the difference should shrink by ≈4; for Simpson by ≈16.
- **Richardson extrapolation**: Combine I(h) and I(h/2) to obtain a higher-order estimate and an error indicator.
- **Derivative bound**: If $|f''| \le M$ or $|f^{(4)}| \le M$ on [a,b], plug into the error formula to obtain a rigorous a-priori bound.
- **Consistency check**: Integrate a polynomial of degree ≤ the method's precision; the result must be exact (within rounding).
- **Independent method**: Cross-check Trapezoidal result with Simpson (or vice-versa) on the same nodes when possible.